(FP2) 4. Integration 专题作业1

NO. 1

OR

Let $I_n = \int_1^{\sqrt{2}} (x^2 - 1)^n \, \mathrm{d}x$.

i. Show that, for $n \geq 1$, [5]

$$(2n + 1)I_n = \sqrt{2} - 2nI_{n-1}.$$

.....

ii. Using the substitution $x = \sec\theta$, show that [4]

$$I_n = \int_0^{\frac{1}{4}\pi} \tan^{2n+1}\theta \sec\theta \, \mathrm{d}\theta.$$

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iii. Deduce the exact value of [5]

$$\int_0^{\frac{1}{4}\pi} \frac{\sin^7\theta}{\cos^8\theta} \, \mathrm{d}\theta.$$

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NO. 2

Let $I_n = \int_0^1 (1+3x)^n e^{-3x}\,dx$, where n is an integer.

a. Show that $3I_n = 1 - 4^n e^{-3} + 3nI_{n-1}$. [3]

b. Find the exact value of I_2. [3]

NO. 3

The integral I_n, where n is an integer, is defined by $I_n = \int_0^{\frac{1}{2}} \left(1-x^2\right)^{-\frac{1}{2}n} \mathrm{d}x$.

a. Find the exact value of I_1. [2]

b. By considering $\frac{\mathrm{d}}{\mathrm{d}x}\left(x\left(1-x^2\right)^{-\frac{1}{2}n}\right)$, or otherwise, show that $nI_{n+2} = 2^{n-1}3^{-\frac{1}{2}n} + (n-1)I_n$. [5]

c. Find the exact value of I_5 giving the answer in the form $k\sqrt{3}$, where k is a rational number to be determined. [3]

NO. 4

It is given that, for $n \geq 0$,

$$I_n = \int_0^1 x^n e^{x^3} \, dx.$$

i. Show that $I_2 = \frac{1}{3}(e - 1)$. [2]

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ii. Show that, for $n \geq 3$, [3]

$$3I_n = e - (n - 2)I_{n-3}.$$

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iii. Hence find the exact value of I_8. [3]

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NO. 5

Let $I_n = \int_0^{\frac{1}{4}\pi} \sec^n x \, dx$ for $n > 0$.

i. Find the value of I_2. [2]

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ii. Show that, for $n > 2$, [5]

$$(n-1)I_n = 2^{\frac{1}{2}n-1} + (n-2)I_{n-2}.$$

.....

iii. The curve C has equation $y = \sec^3 x$ for $0 \leq x \leq \frac{1}{4}\pi$. The region R is bounded by C, the x-axis, the y-axis and the line $x = \frac{1}{4}\pi$. Find the volume of revolution generated when R is rotated through 2π radians about the x-axis. [4]

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NO. 6

i. Let $I_n = \int_0^{\frac{1}{2}\pi} \cos^n x \sin^2 x \, dx$, for $n \geq 0$. By differentiating $\cos^{n-1} x \sin^3 x$ with respect to x, prove that [5]

$$(n+2)I_n = (n-1)I_{n-2} \quad \text{for } n \geqslant 2.$$

ii. Hence find the exact value of I_4. [4]

NO. 7

Find the exact value of $\int_2^{\frac{7}{2}} \frac{1}{\sqrt{4x-x^2-1}}\,dx$. [5]

NO. 8

a. Starting from the definitions of tanh and sech in terms of exponentials, prove that

$$1 - \tanh^2 x = \operatorname{sech}^2 x.$$ [3]

b. Using the substitution $u = \tanh x$, or otherwise, find $\int \operatorname{sech}^2 x \tanh^2 x \, dx$. [2]

It is given that, for $n \geqslant 0$, $I_n = \int_0^{\ln 3} \operatorname{sech}^n x \tanh^2 x \, \mathrm{d}x$.

c. Show that, for $n \geqslant 2$,

$$(n+1)I_n = \left(\frac{4}{5}\right)^3 \left(\frac{3}{5}\right)^{n-2} + (n-2)I_{n-2}.$$ [5]

[You may use the result that $\frac{\mathrm{d}}{\mathrm{d}x}(\operatorname{sech} x) = -\tanh x \operatorname{sech} x$.]

d. Find the value of I_4. [3]